



Nivolumab with or without ipilimumab treatment for metastatic sarcoma (Alliance A091401): two open-label, non-comparative, randomised, phase 2 trials

Sandra P D'Angelo, Michelle R Mahoney, Brian A Van Tine, James Atkins, Mohammed M Milhem, Balkrishna N Jahagirdar, Cristina R Antonescu, Elise Horvath, William D Tap, Gary K Schwartz, Howard Streicher

Summary

Background Patients with metastatic sarcoma have limited treatment options. Nivolumab and ipilimumab are monoclonal antibodies targeting PD-1 and CTLA-4, respectively. We investigated the activity and safety of nivolumab alone or in combination with ipilimumab in patients with locally advanced, unresectable, or metastatic sarcoma.

Methods We did a multicentre, open-label, non-comparative, randomised, phase 2 study that enrolled patients aged 18 years or older and had central pathology confirmation of sarcoma with at least one measurable lesion by Response Evaluation Criteria In Solid Tumors (RECIST) 1.1, evidence of metastatic, locally advanced or unresectable disease, an ECOG performance status of 0–1, and received at least one previous line of systemic therapy. Patients were assigned to treatment in an unblinded manner, as this trial was conducted as two independent, non-comparative phase 2 trials. Enrolled patients were assigned (1:1) via a dynamic allocation algorithm to intravenous nivolumab 3 mg/kg every 2 weeks, or nivolumab 3 mg/kg plus ipilimumab 1 mg/kg every 3 weeks for four doses. Thereafter, all patients received nivolumab monotherapy (3 mg/kg) every 2 weeks for up to 2 years. The primary endpoint was the proportion of patients with locally advanced, unresectable or metastatic soft tissue sarcoma achieving a confirmed objective response. Analysis was per protocol. This study is ongoing although enrolment is closed. It is registered with ClinicalTrials.gov, number NCT02500797.

Findings Between Aug 13, 2015, and March 17, 2016, 96 patients from 15 sites in the USA underwent central pathology review for eligibility and 85 eligible patients, including planned over-enrolment, were allocated to receive either nivolumab monotherapy (43 patients) or nivolumab plus ipilimumab (42 patients). The primary endpoint analysis was done according to protocol specifications in the first 76 eligible patients (38 patients per group). The number of confirmed responses was two (5% [92% CI 1–16] of 38 patients) in the nivolumab group and six (16% [7–30] of 38 patients) in the nivolumab plus ipilimumab group. The most common grade 3 or worse adverse events were anaemia (four [10%] patients), decreased lymphocyte count (three [7%]), and dehydration, increased lipase, pain, pleural effusion, respiratory failure, secondary benign neoplasm, and urinary tract obstruction (two [5%] patients each) among the 42 patients in the nivolumab group and anaemia (eight [19%] patients), hypotension (four [10%] patients), and pain and urinary tract infection (three [7%] patients each) among the 42 patients in the nivolumab plus ipilimumab group. Serious treatment-related adverse events occurred in eight (19%) of 42 patients receiving monotherapy and 11 (26%) of 42 patients receiving combination therapy, and included anaemia, anorexia, dehydration, decreased platelet count, diarrhoea, fatigue, fever, increased creatinine, increased alanine aminotransferase, increased aspartate aminotransferase, hyponatraemia, pain, pleural effusion, and pruritus. There were no treatment-related deaths.

Interpretation Nivolumab alone does not warrant further study in an unselected sarcoma population given the limited efficacy. Nivolumab combined with ipilimumab demonstrated promising efficacy in certain sarcoma subtypes, with a manageable safety profile comparable to current available treatment options. The combination therapy met its predefined primary study endpoint; further evaluation of nivolumab plus ipilimumab in a randomised study is warranted.

Funding Alliance Clinical Trials in Oncology, National Cancer Institute Cancer Therapy Evaluation Program, Bristol-Myers Squibb, Cycle for Survival.

Introduction

Sarcomas are rare, heterogeneous malignant tumours of mesenchymal origin characterised into more than 100 distinct subtypes, accounting for 1% of malignancies in adults.¹ For newly diagnosed patients with metastatic sarcoma who are chemotherapy naive, the efficacy of approved treatments, doxorubicin alone or gemcitabine

plus docetaxel, is similar,² with approximately 18% of patients achieving objective responses with these regimens, with progression-free survival of approximately 5 months and overall survival of 16 months. Beyond the front-line setting, the US Food and Drug Administration (FDA) has granted approval for systemic agents, including pazopanib, trabectedin, and eribulin, for

Lancet Oncol 2018

Published Online

January 19, 2018

[http://dx.doi.org/10.1016/](http://dx.doi.org/10.1016/S1470-2045(18)30006-8)

[S1470-2045\(18\)30006-8](http://dx.doi.org/10.1016/S1470-2045(18)30006-8)

See Online/Comment

[http://dx.doi.org/10.1016/](http://dx.doi.org/10.1016/S1470-2045(18)30039-1)

[S1470-2045\(18\)30039-1](http://dx.doi.org/10.1016/S1470-2045(18)30039-1)

Memorial Sloan Kettering Cancer Center, New York, NY, USA (S P D'Angelo MD, Prof C R Antonescu MD, W D Tap MD); Weill Cornell Medical College, New York, NY, USA (S P D'Angelo, W D Tap); Alliance Statistics and Data Center, Mayo Clinic, Rochester, MN, USA (M R Mahoney MD); Washington University in Saint Louis School of Medicine, St Louis, MO, USA (B A Van Tine MD); Southeast Clinical Oncology Research Consortium NCORP, Winston-Salem, NC, USA (J Atkins MD); University of Iowa/Holden Comprehensive Cancer Center, Iowa City, IA, USA (Prof M M Milhem MD); Regions Hospital Cancer Care Center, Saint Paul, MO, USA (B N Jahagirdar MD); University of Minnesota, Minneapolis, MO, USA (B N Jahagirdar); Astellas, Northbrook, IL, USA (E Horvath MD); Alliance for Clinical Trials in Oncology, University of Chicago, Chicago, IL, USA (E Horvath); Columbia University School of Medicine/Herbert Irving Cancer Center, New York, NY, USA (Prof G K Schwartz MD); and National Cancer Institute, Cancer Therapy Evaluation Program, Investigational Drug Branch, Bethesda, MD, USA (H Streicher MD)

Correspondence to:

Dr Sandra P D'Angelo, Memorial Sloan Kettering Cancer Center, New York, NY 10065, USA
dangelos@mskcc.org

Research in context

Evidence before this study

We searched PubMed using the terms, “metastatic sarcoma” and “immunotherapy or immune checkpoint and sarcoma” for articles published up to June 1, 2017, without language restrictions. We excluded review articles and meta-analyses. Programmed death ligand 1 (PD-L1) expression has been investigated in sarcoma samples. Results have varied with the assay used, type of sarcoma, and timing of testing thereby limiting its role as a predictive biomarker of treatment benefit. A comprehensive overview of the sarcoma immune microenvironment was reported for four sarcoma subtypes, including undifferentiated pleomorphic sarcoma, leiomyosarcoma, synovial sarcoma, and liposarcoma. In this analysis, undifferentiated pleomorphic sarcoma and leiomyosarcoma were shown to have high expression of genes related to antigen presentation and T-cell infiltration compared with synovial sarcoma or liposarcoma. These analyses could provide rationale for the exploration of immunotherapeutic approaches, including checkpoint inhibitors, in sarcoma. Thus far, there has been limited success with either T-lymphocyte (CTLA-4) or anti programmed death-1 (PD-1) antibodies, alone in sarcomas such as leiomyosarcoma, synovial sarcoma and most bone sarcomas. However, pembrolizumab led to promising results in undifferentiated pleomorphic sarcoma and liposarcoma.

Added value of this study

To our knowledge, this is the first investigation of combination checkpoint inhibition in patients with sarcoma. Combination treatment with nivolumab plus ipilimumab in patients with advanced, previously treated sarcoma achieved promising objective responses that seemed to be clinically meaningful in undifferentiated pleomorphic sarcoma, leiomyosarcoma, myxofibrosarcoma, and angiosarcoma. Additionally, the current dose and schedule for the combination tested (3 mg/kg nivolumab plus 1 mg/kg ipilimumab) had acceptable toxicity, with 14% of patients having grade 3 or 4 treatment-related adverse events. Our results are similar to those obtained with existing systemic chemotherapeutic agents. The combination cohort met the primary endpoint, thereby forming the basis for a confirmatory phase 3 clinical trial that is being planned in selected sarcoma subtypes in which responses were observed.

Implications of all the available evidence

Monotherapy checkpoint inhibition has promising activity in undifferentiated pleomorphic sarcoma and liposarcoma. Our findings might support future studies of nivolumab plus ipilimumab in patients with these specific subtypes of metastatic sarcoma, as well as those with angiosarcoma, leiomyosarcoma, and myxofibrosarcoma. These findings also highlight the need for the identification of a predictive biomarker that can inform treatment decisions.

selected sarcoma subtypes.^{3–5} Each of these agents conferred modest improvements in either progression-free survival or overall survival. Yet the overall proportion of patients achieving objective responses remains less than 10%, with median progression-free survival of 4 months and overall survival of less than 14 months.^{3–5} In a recent phase 2 study in patients with advanced soft tissue sarcoma,⁶ the median overall survival of patients treated with doxorubicin plus olaratumab was superior to the monotherapy (stratified hazard ratio [HR] 0.46, 95% CI 0.30–0.71, $p=0.0003$), 26.5 months (20.9–31.7) versus 14.7 months (9.2–17.1) in the doxorubicin alone group, which lead to the approval of doxorubicin plus olaratumab treatment in these patients.⁶ These findings are pending confirmation in a larger, randomised, phase 3 study (NCT02451943), which has now closed to accrual. Despite these recent approvals by the FDA, the need for therapies that are less toxic, and offer durable disease control and improved survival in patients with metastatic sarcoma remains.

Modulating the immune system with monoclonal antibodies that block immune checkpoints has emerged as a promising strategy in cancer care that leads to durable antitumour activity and improved survival in multiple malignancies when compared with other systemic therapies.^{7–13} Immune checkpoints molecules such as programmed death ligand 1 (PD-L1) can be

overexpressed by tumours or the tumour micro-environment, inhibiting the antitumour activity of effector T cells.¹⁴ Some sarcomas express PD-L1, with reported expression of positive PD-L1 cells ranging from 12% to 65%, depending on the sarcoma histology, timing of sample collection, and assay used for PD-L1 detection.^{15,16} This variability poses a challenge in the use of PD-L1 expression as a prognostic biomarker. Additionally, clinical data from patients treated with checkpoint inhibitors have suggested that patients can benefit from this therapy irrespective of PD-L1 expression.^{8,9,12,17} While this finding highlights the need for more effective biomarkers, it should not preclude the exploration of available checkpoint inhibitors in sarcoma.

Limited evidence exists on the activity of checkpoint inhibitors in sarcoma. In a small phase 2 study,¹⁸ six patients with synovial sarcoma were treated with ipilimumab 3 mg/kg every 3 weeks without achieving any documented responses. Another phase 2 study¹⁹ in 80 patients with bone or soft tissue sarcomas showed that 18% of patients in the soft tissue sarcoma group achieved objective responses, with all these patients having either undifferentiated pleomorphic sarcoma or liposarcoma. Of the 40 patients with bone sarcoma, only two had objective responses, one in a patient with osteosarcoma and the other in a patient with dedifferentiated chondrosarcoma. The low clinical

activity of pembrolizumab alone in most sarcoma subtypes suggests that this agent cannot adequately activate suppressed effector T cells in these patients. Undifferentiated pleomorphic sarcoma is an inflamed tumour characterised by high numbers of tumour infiltrating lymphocytes, which could explain the clinical activity of pembrolizumab noted in this subgroup of sarcoma patients.²⁰ Most subtypes of sarcoma do not have infiltration of immune cells, emphasising the need to explore other combinatorial immunotherapy strategies. Combination of agents that enhance anti-tumour effects through different mechanisms, such as cytotoxic T-lymphocyte (CTLA-4) and programmed death-1 (PD-1), have been shown to be synergistic in preclinical mouse models of melanoma.²¹ Additionally, clinical data of these agents in multiple malignancies, have shown their efficacy when compared with chemotherapy regimens.^{7,10,22} We did this study to assess the activity of nivolumab alone and nivolumab plus ipilimumab in patients with metastatic sarcoma.

Methods

Study design and participants

This multicentre, open-label, non-comparative, randomised, phase 2 trial recruited patients from 15 centres in the USA that were members of the Alliance Clinical Trials in Oncology Group (A091401) and the National Clinical Trials Network (NCTN; appendix p 11).

Eligible patients aged 18 years or older had histologically confirmed bone or soft tissue sarcoma by central pathology review, locally advanced, unresectable, or metastatic sarcoma by Response Evaluation Criteria In Solid Tumors (RECIST) version 1.1 with at least one measurable lesion, and an Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. Enrolled patients had an estimated life expectancy of 3 months and needed to have received at least one previous systemic therapy for sarcoma, with a minimum washout period of 28 days before the initiation of the study drugs. Previous therapy with any anti-PD-1, anti-PD-L1, or anti-CTLA-4 blocking antibodies was not permitted. Disease progression was not an inclusion criteria. Key exclusion criteria included active brain metastases or a history of autoimmune diseases. Patients also needed adequate kidney, liver, and bone marrow function. Full eligibility criteria including other exclusion criteria and the required initial laboratory values are provided in the appendix (pp 1, 2).

Eligibility criteria for patients who were allowed to crossover due to disease progression from the nivolumab monotherapy group to the nivolumab plus ipilimumab group included radiographical or clinical confirmation of disease progression, a minimum of 10 weeks of single agent nivolumab treatment, and resolution of any toxic effects of previous therapy except fatigue and alopecia. All crossover eligibility criteria are detailed in the appendix (p 4).

Review boards at each of the participating institutions approved the study. The study was done in accordance with the Declaration of Helsinki and the Guidelines for Good Clinical Practice. Each participant signed an institutional review board-approved, protocol-specific, informed consent form in accordance with federal and institutional guidelines. This study was monitored at least twice per year by the Data and Safety Monitoring Board, a standing committee consisting of individuals outside of the Alliance Clinical Trials in Oncology Group for accrual, safety, and the primary endpoint evaluation.

Randomisation and masking

Patients were assigned to treatment in an unblinded manner because this trial was done as two independent, non-comparative, phase 2 trials. Following registration, each patient was assigned one of the two treatments (1:1) by use of a dynamic allocation algorithm based on the methods by Pocock and Simon.²³ The assigned treatment was automatically generated by systems within the Alliance Clinical Trials in Oncology Group, Statistics and Data Center and immediately transferred to the appropriate system in each institution for assignment of the patient. Manual lists of the assignments were not generated. Site of randomisation was the only stratification factor.

Procedures

Patients received either nivolumab 3 mg/kg as an intravenous infusion every 2 weeks until disease progression for a maximum of 2 years after registration or nivolumab 3 mg/kg and ipilimumab 1 mg/kg every 3 weeks for four doses followed by nivolumab 3 mg/kg every 2 weeks until disease progression or a maximum of 2 years after registration. Dose reductions were not permitted; however, dose interruptions for up to 6 weeks were allowed. If treatment was discontinued because of adverse events, patients were followed up until disease progression, initiation of different therapy, or 30 days after the last dose of nivolumab or nivolumab plus ipilimumab.

Tumour assessments were done by radiological assessment at each institution at baseline, every 6 weeks for the first 12 weeks, and every 8 weeks thereafter. Responses detected at these assessments required confirmation 4 weeks after the initial response. Central radiology review was not done. The activity analyses were done per protocol, including all patients who started treatment. Patients who had progressive disease within the first 12 weeks of treatment were permitted to continue treatment beyond the initial progressive disease if the following prespecified criteria were met: evidence of clinical benefit, tolerance of study treatment, no more than four new lesions, and less than a 40% increase in tumour burden. If progressive disease was confirmed at the subsequent 4 week assessment, the date of the initial progressive disease was used for analyses and the patient stopped study treatment. Otherwise, if progressive

See Online for appendix

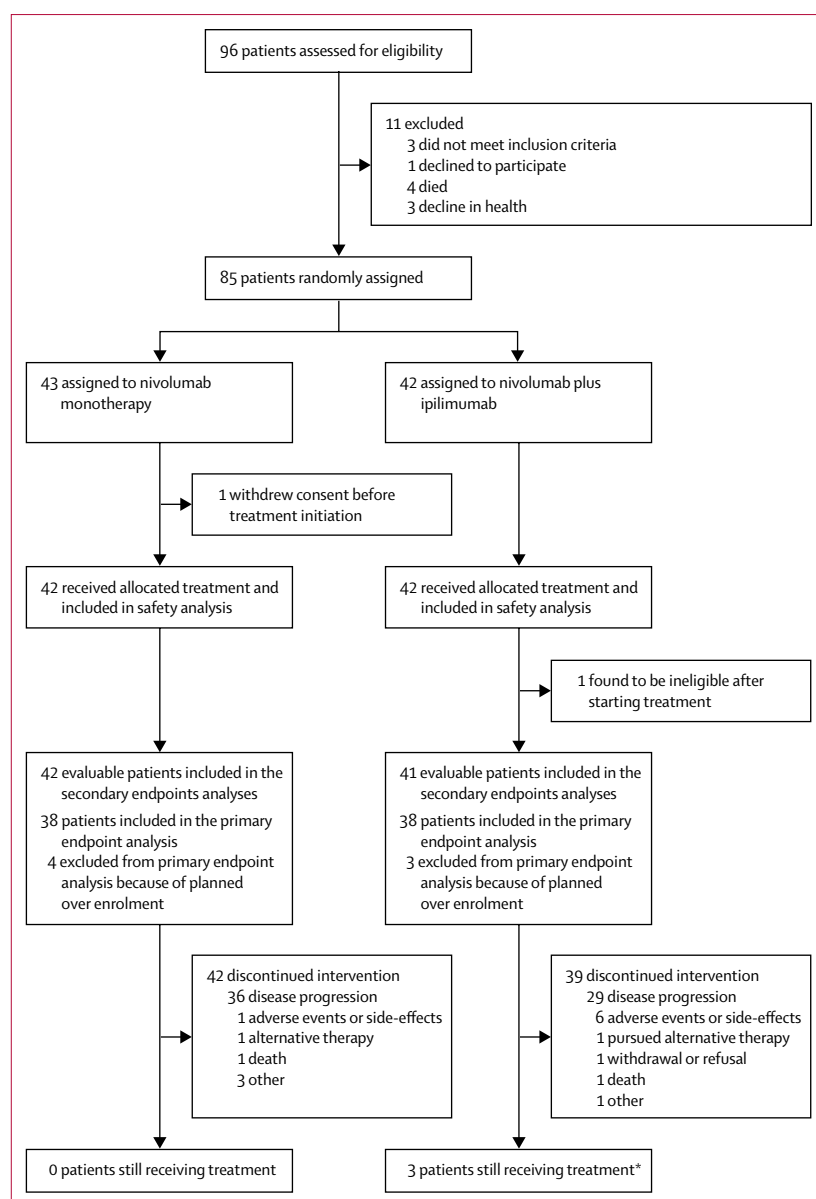


Figure 1: Trial profile

*Patients still on treatment as of the data lock date of April 24, 2017.

disease was not confirmed, the patient could remain on protocol treatment and the prior scan where progressive disease was suspected was used, going forward, as the baseline for determining tumour response. 12 months after the trial opened to enrolment, a protocol amendment allowed patients who were receiving nivolumab monotherapy and had progressive disease to crossover to the nivolumab plus ipilimumab group provided that eligibility criteria were met (appendix p 4). Data acquired after crossover was not included in the evaluation of the primary endpoint.

Safety analyses were done by including all randomised patients who began treatment. Laboratory monitoring

and safety assessments were done at baseline, and every 2–3 weeks as per the treatment schedule. Adverse events were graded in accordance with the National Cancer Institute Common Terminology Criteria for Adverse Events version 4.0 during treatment and up to 30 days after treatment discontinuation.

After discontinuation of protocol treatment, data on survival, disease status, first non-protocol systemic therapy, and long-term adverse events were captured every 6 months for a maximum of 3 years after randomisation.

Optional biopsies before the first treatment and at week 6 of treatment were obtained from patients who provided informed consent. Additionally, blood samples for prespecified exploratory analyses were obtained at screening and selected timepoints during study treatment.

All analyses excluded any newly collected data for patients who withdrew consent from future follow-up and beyond the date of withdrawal.

Outcomes

The primary endpoint was confirmed objective response, defined as complete or partial response by RECIST version 1.1 on two consecutive evaluations at least 4 weeks apart during the protocol-directed treatment. The proportion of confirmed objective responses was estimated as the number of patients with a complete response or partial response on two consecutive evaluations at least 4 weeks apart, divided by the number of evaluable patients.

Secondary endpoints were duration of response, the proportion of patients achieving a clinical benefit, progression-free survival, and overall survival. The clinical benefit was defined as the proportion of patients with a best objective status of complete response, partial response, or stable disease at a given timepoint while receiving protocol treatment. Duration of response was defined as the time from first complete response or partial response to date of progressive disease or death. Progression-free survival was defined as the time from randomisation to date of progressive disease or death. Patients who discontinued treatment for reasons other than progressive disease (such as adverse events) were censored at the date of their most recent disease evaluation prior to receiving any future systemic treatment regimens. Overall survival was calculated as the time between randomisation and date of death. Patients who were lost to follow-up were censored for survival at the date they were last known to be alive and with progressive disease at the date of their most recent disease assessment.

Statistical analysis

For each treatment regimen, the original study design was a two stage Simon stage 2 clinical trial to assess whether the confirmed response rate was at most 5% against the alternative that the confirmed response rate was at least 20%. The rapid registration led to the study design being converted (protocol update 1, dated Nov 15,

2015) to a single stage design with same hypotheses, significance level, and power following consensus from the study team, the Alliance Data and Safety Monitoring Board, the Alliance Experimental Therapeutics and Rare Tumors Committee leadership, and the National Cancer Institute Cancer Therapy Evaluation Program. No outcome data were reviewed during the development of this amendment. Specifically, 38 patients were planned to be enrolled on to each treatment regimen. Five (13%) of these 38 evaluable patients achieving a confirmed response would be considered sufficient evidence of promising clinical activity in this setting. This study design yielded 92% power to detect clinical activity if at least 20% of patients had a confirmed response at 0.04 level of significance (one-sided test), and clinical inactivity if 5% or fewer patients achieved a confirmed response.

92% two-sided binomial CIs are reported alongside the observed proportion of patients achieving a confirmed response within each group.

All enrolled patients who began study treatment were considered evaluable for the analysis of adverse events, including data up to the date of ineligibility for any patients found ineligible after initiating study treatment. Categorical data analyses and summary statistics were used to report adverse events.

We used Kaplan-Meier methods to estimate the distributions of all time-to-event endpoints.²⁵ 95% CIs for statistical estimates were calculated using the natural logarithm (ln) transformation approach. All patients included in the evaluation of the primary endpoint were also used for the evaluation of the time-to-event endpoints.

All statistical analyses were done with SAS version 9.4. This trial is registered with ClinicalTrials.gov, number NCT02500797.

Role of the funding source

Alliance and NCI-CTEP supported the study design, and Alliance Statistics and Data Center did the statistical analyses. Bristol-Myers Squibb did not contribute to study design, data collection, data analysis, data interpretation, or writing of the report, but did review the manuscript. SPD'A and MRM, had full access to all the data in the study, and the corresponding author (SPD'A) had final responsibility for the decision to submit for publication.

Results

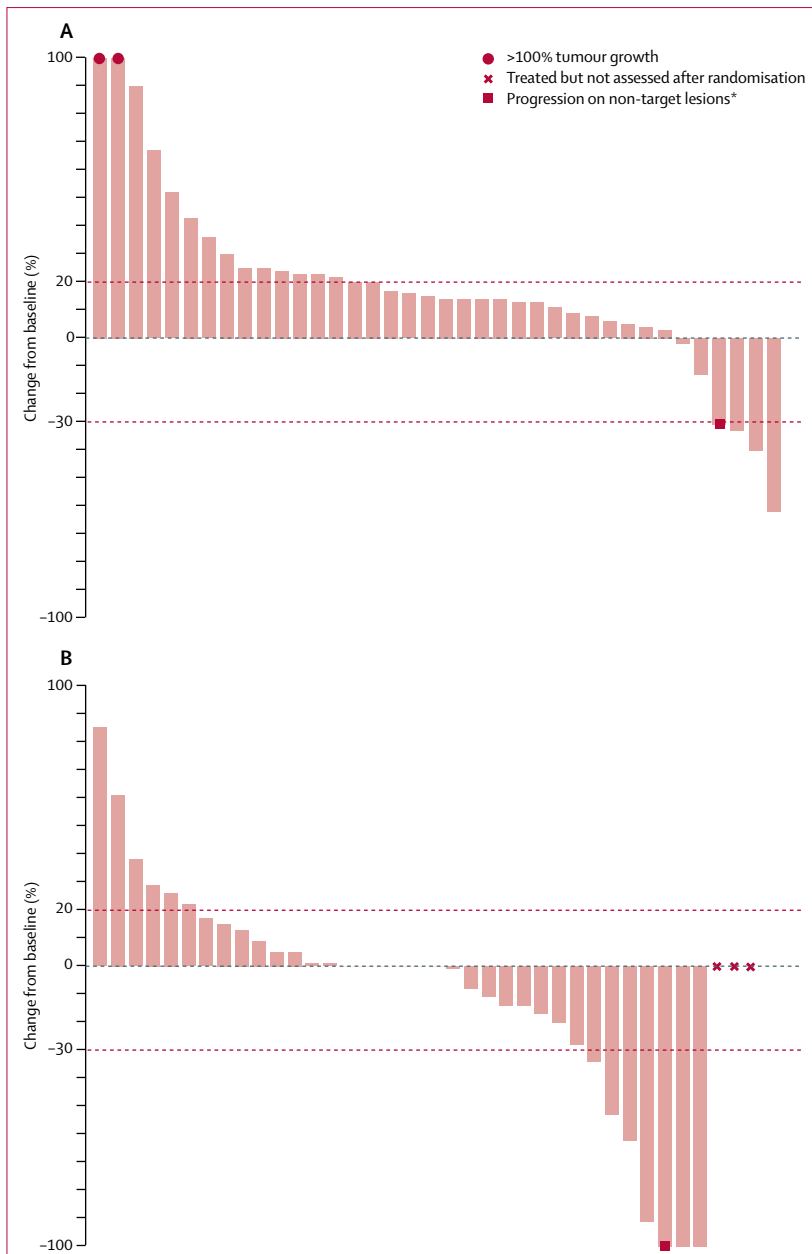
96 patients underwent central pathology review for eligibility, 81 patients from Aug 13 to Dec 24, 2015, and 15 patients from March 16 to March 17, 2016. 11 of 96 patients were excluded, so 85 patients from 13 Alliance Clinical Trials in Oncology sites and two NCTN groups in the USA proceeded to randomisation (appendix p 11). 43 patients were assigned to nivolumab alone group and 42 patients to nivolumab plus ipilimumab group. One

	Nivolumab group (n=43)	Nivolumab plus ipilimumab group (n=42)
Age (years)		
Mean (SD)	52.9 (13.8)	54.1 (13.3)
Median (Range)	56.0 (21.0–76.0)	57.0 (27.0–81.0)
ECOG performance status		
0	28 (65%)	24 (57%)
1	15 (35%)	18 (43%)
Sex		
Female	21 (49%)	23 (55%)
Male	22 (51%)	19 (45%)
Previous therapies		
1	5 (12%)	10 (24%)
2	12 (28%)	6 (14%)
≥3	26 (60%)	26 (62%)
Histological type*		
Angiosarcoma	0	3 (7%)
Bone	5 (12%)†	4 (10%)‡
Leiomyosarcoma	15 (35%)	14 (33%)
Liposarcoma (well differentiated or dedifferentiated)	3 (7%)	2 (5%)
Sarcoma (not otherwise specified)	2 (5%)	1 (2%)
Spindle cell sarcoma	5 (12%)	6 (14%)
Synovial sarcoma	2 (5%)	2 (5%)
Undifferentiated pleomorphic sarcoma/malignant fibrous histiocytoma	5 (12%)	6 (14%)
Other	6 (14%)†	4 (10%)‡
Histological grade (differentiation)		
G1 (well differentiated)	3 (7%)	2 (5%)
G2 (moderately differentiated)	9 (21%)	7 (17%)
G3 (poorly differentiated)	21 (49%)	20 (48%)
GX (grade cannot be assessed)	10 (23%)	13 (31%)

Data are n (%) unless noted otherwise. ECOG=Eastern Cooperative Oncology Group. *Based on a Central Review of pathology, prior to randomisation. †Bone or other sarcomas in the nivolumab group consisted of three patients with Ewing's sarcoma and one patient each with dedifferentiated chondrosarcoma, osteogenic sarcoma, alveolar soft part sarcoma, epithelioid sarcoma, extraskeletal myxoid chondrosarcoma, malignant solitary fibrous tumour, malignant peripheral nerve sheath tumour, and neoplasm with perivascular epithelioid cell differentiation. ‡Bone or other sarcomas in the nivolumab plus ipilimumab group consisted of two patients with Ewing's sarcoma and one patient each with dedifferentiated chondrosarcoma, osteosarcoma, alveolar soft part sarcoma, extraskeletal myxoid chondrosarcoma, malignant peripheral nerve sheath tumour, and myxofibrosarcoma.

Table 1: Baseline demographics and disease characteristics for all randomised patients

patient withdrew consent to initiate nivolumab monotherapy after randomisation and was excluded from the safety and efficacy analyses. Another patient was deemed to be ineligible after starting combination therapy because of non-evaluable disease and was excluded from the primary endpoint analysis. Safety analyses included all patients who received study treatment, 42 patients in each group. Secondary outcome analyses are presented for all evaluable



Figures 2: Maximum percentage change in tumour size by treatment group
Waterfall plots show the percentage change in the sum of the target lesions or nodes from study entry to confirmatory disease progression assessment after treatment initiation for each patient during active (A) nivolumab monotherapy or (B) nivolumab plus ipilimumab combination therapy. Horizontal dashed lines represent the criteria for progressive disease (20% increase in tumour size) and response (30% decrease in tumour size for partial response; total disappearance of target lesions or nodes for complete response), as per RECIST version 1.1. RECIST=Response Evaluation Criteria In Solid Tumors. *Patient achieved partial response according to radiographic assessments, but was classified as progressive disease by unequivocal progressive disease on non-target lesions (as per RECIST version 1.1 requirements).

patients: 42 patients in the monotherapy group and 41 patients in the combination group. Efficacy data for the primary endpoint are presented for the first 38 evaluable patients in each group, as specified in the protocol (figure 1).

Patient demographics and disease characteristics at baseline are summarised in table 1 for the two treatment groups. Patients were heavily pretreated, with 52 (61%) of 85 patients having received at least three previous lines of chemotherapy. The most common sarcoma types across both groups were bone sarcoma (nine [11%] patients), leiomyosarcoma (29 [34%] patients), liposarcoma (five [6%] patients), spindle cell sarcoma (11 [13%] patients), and undifferentiated pleomorphic sarcoma or malignant fibrous histiocytoma (11 [13%] patients; table 1).

Data collection was locked as of April 24, 2017. In the nivolumab monotherapy group, 15 (32%) of 42 evaluable patients were alive at the time of analysis, with a median follow-up of 13·6 months (IQR 8·9–15·88) and a median duration of treatment of 2·3 months (1·5–3·5). 27 patients died; 25 from their disease and two for unknown reasons. Two of the 38 evaluable patients (one with alveolar soft part sarcoma and the other with non-uterine leiomyosarcoma) had a confirmed partial response. Thus, the proportion of patients achieving a confirmed response rate was 5% (92% CI 1–16). One patient with sarcoma, not otherwise specified, had an unconfirmed partial response (figure 2A; appendix p 6). Four additional patients were also treated but were non-evaluable for the primary per protocol analysis, none of whom had a response.

In the combination group, 20 (49%) of 41 evaluable patients were alive at the time of analysis, with a median follow-up of 14·2 months (IQR 12·7–16·1) and a median duration of treatment of 3·7 months (2·1–4·6). 21 patients died; 16 from their disease, four for unknown reasons, and one from sepsis. Six of the 38 evaluable patients had a confirmed response. Thus, the proportion of patients achieving a confirmed response rate was 16% (92% CI 7–30). Confirmed responses were seen in patients with uterine leiomyosarcoma (n=1), non-uterine leiomyosarcoma (n=1), myxofibrosarcoma (n=1), undifferentiated pleomorphic sarcoma or malignant fibrous histiocytoma (n=2), and angiosarcoma (n=1). One patient with undifferentiated pleomorphic sarcoma or malignant fibrous histiocytoma had an unconfirmed response (figure 2B; appendix p 6). Three additional patients were treated but were non-evaluable for the primary per protocol analysis, none of whom had responses.

In the monotherapy group, the two confirmed responses were first detected 1·2 months and 1·3 months after initiating treatment and lasted for 3·2 months and 11·6 months, respectively (figure 3A). In the combination group, the six confirmed responses were first detected 1·3–4·4 months after initiating treatment with a median duration of response of 6·2 months (range 1·4 to >14·1; figure 3B).

In the monotherapy group, 29 (69% of 42) patients had progressive disease at week 12, of whom 18 (62%) were eligible to continue treatment. 11 (61%) of these 18 patients were confirmed as having progressive disease

within 1 month, and seven (39%) continued receiving treatment for 2.0 to 8.5 months. One of the 11 patients confirmed as having progressive disease within 1 month continued treatment in error for an additional 3 weeks, ending treatment at that time because of a third disease status of progressive disease (appendix p 7). In the combination therapy group, 18 (43% of 42) patients had progressive disease within the first 12 weeks, of whom eight (44%) were eligible to continue treatment. Three (38%) of these eight patients had progressive disease at confirmation. The other five (62%) patients had stable disease at confirmation and remained on treatment for another 3–12 months. One patient converted to a partial response from progression within 3 months of initial progressive disease, which lasted for 9.2 months (appendix p 8).

In the monotherapy group, clinical benefit was shown by four (10%; 95% CI 3–22) of 42 patients 6 months after treatment initiation and one patient (2%, 1–12) at 12 months. 37 (88%) of 42 evaluable patients had progressive disease. In the combination group, two responses were ongoing as of the time of analysis, two cases of progressive diseases occurred, and three patients ended protocol treatment because of adverse events (figure 3B). In the combination group, clinical benefit was shown by six (12%, 6–28) of 42 patients 6 months after treatment initiation and five (12%, 5–25) patients at 12 months.

Median progression-free survival was 1.7 months (95% CI 1.4–4.3; n=42; figure 4A) in the monotherapy group and 4.1 months (2.6–4.7; n=41; figure 4B) in the combination group.

Median overall survival was 10.7 months (5.5–15.4; n=42; figure 4A) in the monotherapy group and 14.3 months (9.6–not reached; n=41; figure 4B) in the combination group.

After the protocol amendment was approved, four (19%) of 21 eligible patients in the nivolumab monotherapy group crossed over to the combination therapy group and developed progressive disease within 0.4–2.8 months.

Adverse events of grade 1–2 observed in 10% or more patients and all grade 3, 4 and 5 events are reported in table 2. All of the 84 patients (42 in each group) who initiated treatment had adverse events of any grade. In the monotherapy group, the most common grade 3 or worse adverse events were anaemia (four [10%] patients), decreased lymphocyte count (three [7%]), and dehydration, increased lipase, pain, pleural effusion, respiratory failure, secondary benign neoplasm, and urinary tract obstruction (two [5%] patients each). In the combination group, the most common grade 3 or worse adverse events were anaemia (eight [19%] patients), hypotension (four [10%] patients), and pain and urinary tract infection (three [7%] patients each). The incidence of treatment-related adverse events of any grade was similar in both treatment groups. The most common

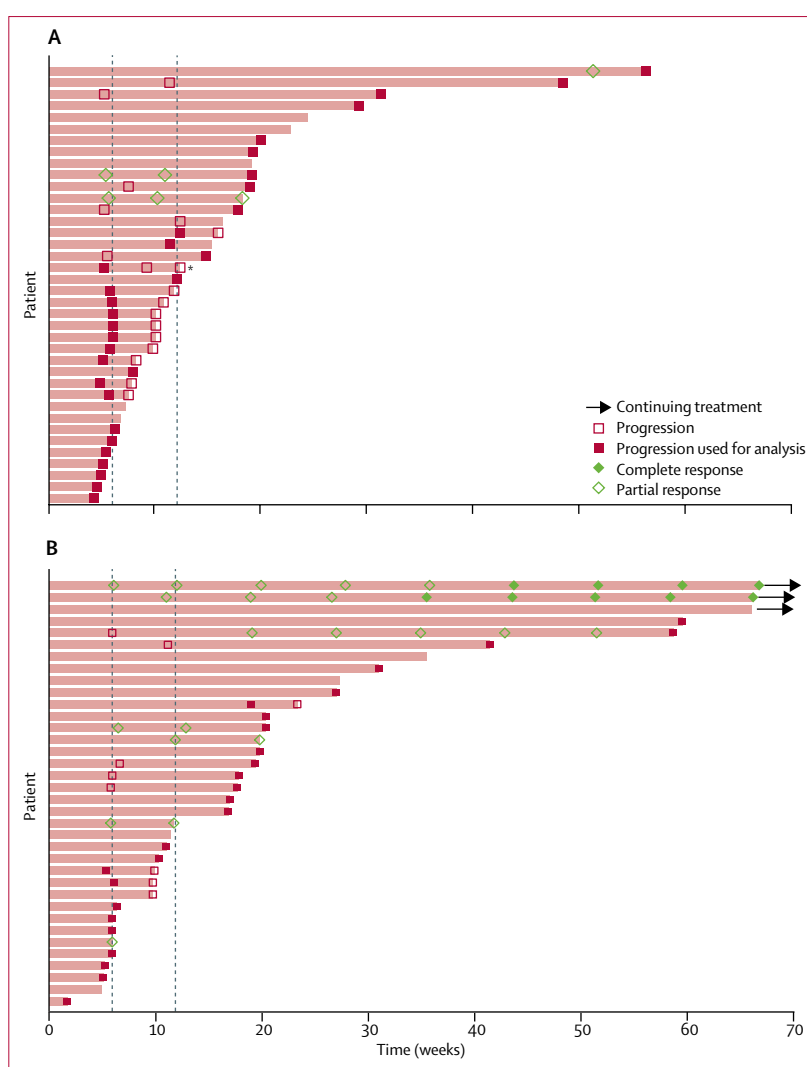
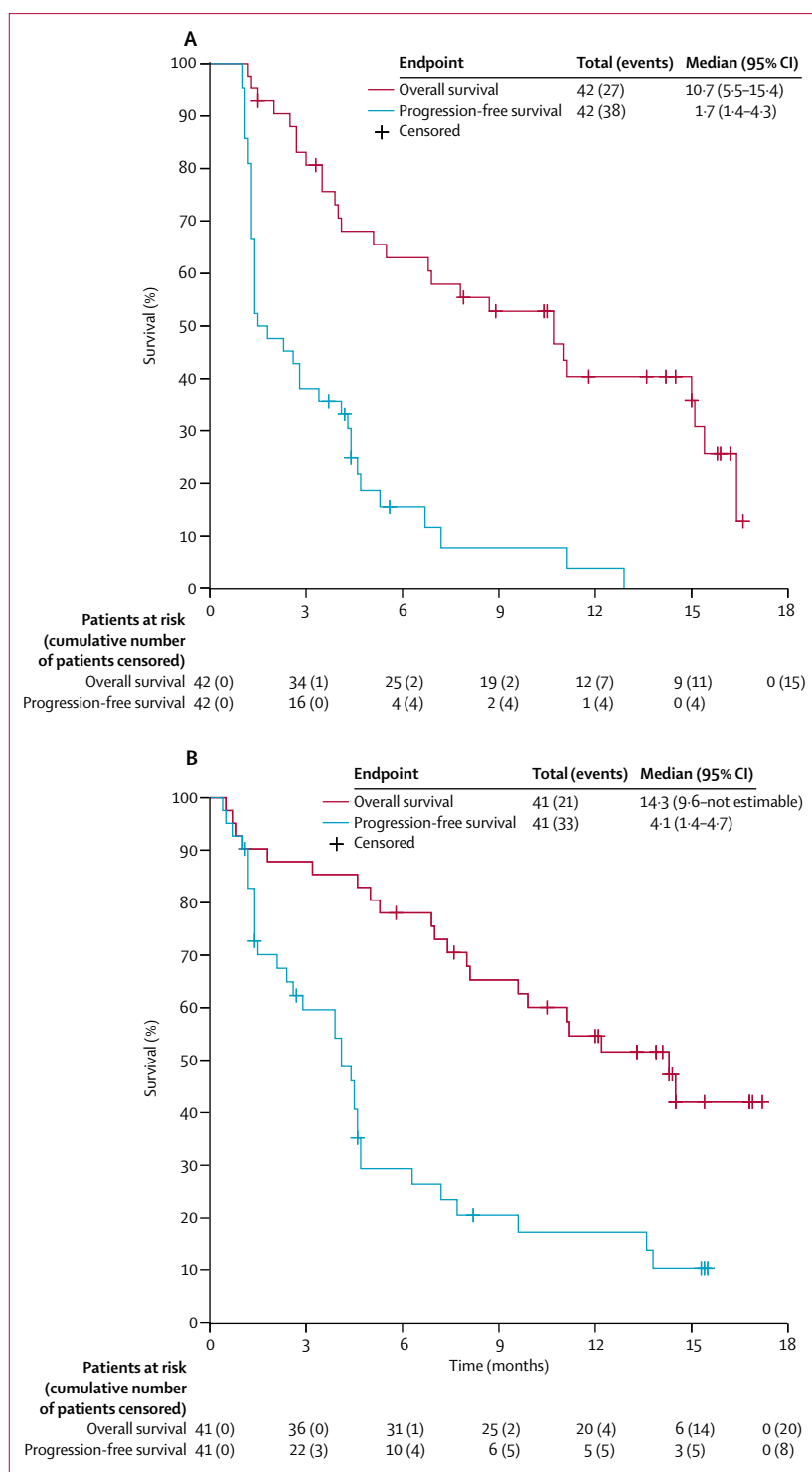


Figure 3: Over time tumour response by treatment

Swimmer plots have one lane for each patient's initial treatment period (from study entry to confirmatory disease progression assessment after treatment initiation) during (A) nivolumab monotherapy or (B) nivolumab plus ipilimumab combination therapy. Vertical dashed lines represent the 6 week and 12 week restaging. Squares, empty or filled in, represent suspected and confirmed progressive disease, respectively. An arrow at the end of a lane represent a patient who was actively receiving study treatment at the time of the analysis. Objective responses are represented by diamonds. *One patient had three disease assessments within the first 12 weeks; they had confirmed progressive disease at the second assessment, but continued treatment for an additional 3 weeks and subsequently ended treatment because of a third positive assessment of progressive disease.

treatment-related adverse events in the nivolumab monotherapy group were fatigue (12 [29%] of 42 patients), pain (nine [21%] patients), and anaemia and anorexia (six [14%] patients each). In the combination group, common treatment-related adverse events included fatigue (14 [33%] of 42 patients), skin rash (eight [19%] patients), and adrenal insufficiency, hypothyroidism, pain, and anaemia (six [14%] patients each). Grade 3–4 treatment-related adverse events occurred more often with the combination therapy (six [14%] of 42 patients) than with the monotherapy (three [7%] of 42 patients; appendix p 9). Treatment-related serious adverse events



Figures 4: Survival outcomes by treatment

Kaplan-Meier estimates show the distribution of progression-free survival and overall survival over time with (A) nivolumab monotherapy or (B) nivolumab plus ipilimumab combination therapy. Patients who remained alive or without progression were censored.

occurred in eight (19%) of 42 patients in the monotherapy group, including anaemia, anorexia, dehydration, decreased platelet count, diarrhoea, fever, increased creatinine, and pleural effusion (one [2%] patient each); and in 11 patients (26%) in the combination group, including three (7%) with adrenal insufficiency, two (5%) alanine aminotransferase, two (5%) hyponatraemia, and one (2%) patient each with anaemia, increased aspartate aminotransferase, fatigue, pain, and pruritus (appendix p 10). No drug-related deaths occurred in either group.

At the time of this analysis, all patients in the nivolumab group had discontinued study treatment and only three patients remained on study treatment on the nivolumab plus ipilimumab group. The most common reason for discontinuing study treatment was disease progression (36 patients in the monotherapy group and 29 in the combination group). Adverse events leading to discontinuation of treatment occurred in six (14%) of 42 patients in the combination group and one (2%) of 42 patients in the monotherapy group. In the monotherapy group, nine (21%) patients had dose delays and 22 (52%) had missed doses; in the combination arm, 16 (38%) patients had dose delays and 19 (45%) had dose omissions.

11 patients died of causes secondary to progressive disease during treatment. Five (12%) of 42 patients in the monotherapy group died due to respiratory failure (two patients), thromboembolic events (one patient), and disease progression (two patients). In the combination group, six (14%) of 42 patients died due to respiratory failure (two patients), disease progression (two patients), sepsis (one patient), and dyspnoea (one patient). At the time of the data lock, 27 patients had died in the monotherapy group, 25 patients due to disease progression and two due to unknown causes; 21 patients died in the combination group, with reasons for death being disease progression in 16 patients, unknown in four patients, and sepsis in one patient. No treatment-related deaths occurred in either group.

Discussion

In these phase 2 trials of nivolumab with or without ipilimumab, 85 patients were rapidly enrolled in 5 months. Two (5%) of 38 evaluable patients with advanced sarcoma achieved a confirmed objective response when treated with nivolumab monotherapy, while six (16%) of 38 achieved a confirmed objective response with a combination of nivolumab plus ipilimumab. Across both treatment groups, patients were heavily treated, with 52 (61%) of 85 patients having received at least three previous lines of chemotherapy. Median progression-free survival was 1.7 months (95% CI 1.4-4.3) with monotherapy and 4.1 months (2.6-4.7) with combination therapy. These findings are clinically meaningful when placed in the context of currently available treatment options for patients with metastatic sarcoma. Standard treatments for sarcoma remain

mostly centred around cytotoxic chemotherapy, with front-line therapy typically achieving objective responses in 15–18% of patients, with a median progression-free survival of 4–6 months.² Activity beyond front-line therapy is not improved, with less than 10% of patients achieving objective responses and a median progression-free survival of 2–4 months.^{3–5}

The clinical benefit of nivolumab monotherapy was not on par with currently available chemotherapy-based treatment options. Additionally, the nivolumab monotherapy group did not meet the predefined primary endpoint of achieving responses in at least 13% of patients to prove activity in this setting, which might preclude further study in unselected patients with sarcoma. The combination of nivolumab plus ipilimumab met this predefined primary endpoint target in this population of heavily-treated, unselected, patients with sarcoma. The proportion of patients achieving an objective response with combination therapy was 16%, similar to that achieved with standard chemotherapy-based treatments, such as doxorubicin or gemcitabine plus docetaxel. Additionally, an objective response of 16% in 38 patients is comparable to recently FDA-approved front-line systemic chemotherapy options, potentially supporting further study of nivolumab plus ipilimumab not only as a second-line treatment in patients with sarcoma, but also as a first-line therapy alternative.

In this trial, the median overall survival of patients receiving combination therapy was 14·3 months (95% CI 9·6–not reached). The median overall survival described in similar populations of patients with sarcoma treated with chemotherapy regimens is 11–13·5 months.^{3–5} Although patients treated with the recently FDA-approved combination of doxorubicin plus olaratumab showed a median overall survival of 24 months, these data are pending confirmation in randomised phase 3 clinical trials.⁶ The overall survival observed with nivolumab plus ipilimumab in this clinical trial is encouraging, and suggests potential to improve on the current standard of care for patients with advanced sarcoma.

The safety and tolerability results of these study drugs were consistent with expectations from extensive previous reports.^{9,26,27} In our study, nivolumab monotherapy was generally better tolerated, with lower incidence of adverse events than the combination with ipilimumab. The incidence of treatment-related grade 3–4 adverse events in patients treated with the combination therapy was 14% (six of 42 patients). This finding contrasts with the results of another report of the combination, where the dose of ipilimumab was higher than 3 mg/kg, and 50% of patients had grade 3–4 treatment-related adverse events.²⁷ Using a lower dose of ipilimumab of 1 mg/kg might reduce the frequency of adverse events and make this combinatorial treatment approach safer. Notably, the proportion of grade 3,

	Nivolumab group (n=42)				Nivolumab plus ipilimumab group(n=42)			
	Grade 1–2	Grade 3	Grade 4	Grade 5	Grade 1–2	Grade 3	Grade 4	Grade 5
Adrenal insufficiency	..	..	..	..	5 (12%)	1 (2%)	..	..
Alanine aminotransferase increased	..	..	..	..	..	2 (5%)	1 (2%)	..
Alkaline phosphatase increased	..	..	..	..	..	1 (2%)	..	..
Anaemia	8 (19%)	4 (10%)	..	..	6 (14%)	7 (17%)	1 (2%)	..
Anorexia	10 (24%)	1 (2%)	..	..	14 (33%)	..	..	..
Aspartate aminotransferase increased	..	..	..	..	..	1 (2%)	1 (2%)	..
Cardiac disorders	..	..	..	..	..	1 (2%)	..	..
Colonic perforation	..	..	1 (2%)	..	..	..	..	..
Constipation	5 (12%)	..	1 (2%)	..	6 (14%)	..	..	..
Cough	13 (31%)	..	..	..	8 (19%)	..	..	..
Creatinine increased	..	1 (2%)	..	..	..	1 (2%)	..	..
Dehydration	..	2 (5%)	..	..	..	1 (2%)	..	..
Diarrhoea	4 (10%)	1 (2%)	..	..	11 (26%)	..	..	..
Dyspnoea	10 (24%)	..	..	..	8 (19%)	2 (5%)	..	1 (2%)
Oedema of the limbs	..	..	..	..	..	1 (2%)	..	..
Fatigue	24 (57%)	1 (2%)	..	..	29 (69%)	2 (5%)	..	..
Febrile neutropenia	..	..	1 (2%)	..	..	..	..	..
Fever	4 (10%)	..	..	..	..	..	..	..
Gastrointestinal fistula	..	..	..	..	..	1 (2%)	..	..
Generalised muscle weakness	..	1 (2%)	..	..	..	..	..	..
Genital oedema	..	..	..	..	..	1 (2%)	..	..
Hyperglycaemia	..	..	..	..	..	3 (7%)	..	..
Hyperkalaemia	..	..	..	..	..	2 (5%)	..	..
Hypoalbuminaemia	..	..	..	..	..	2 (5%)	..	..
Hypocalcaemia	..	1 (2%)	..	..	..	..	..	..
Hypokalaemia	..	1 (2%)	..	..	..	1 (2%)	..	..
Hyponatraemia	..	1 (2%)	..	..	..	3 (7%)	1 (2%)	..
Hypophosphataemia	..	1 (2%)	..	..	..	..	..	..
Hypotension	..	..	..	..	..	4 (10%)	..	..
Hypothyroidism	6 (14%)	..	..	..	7 (17%)	..	..	..
Hypoxia	..	..	..	..	..	1 (2%)	..	..
Intraoperative urinary injury	..	1 (2%)	..	..	..	..	..	..
Jejunal obstruction	..	1 (2%)	..	..	..	..	..	..
Leucocytosis	..	..	..	..	..	1 (2%)	..	..
Lipases increased	..	2 (5%)	..	..	..	2 (5%)	..	..
Localised oedema	..	..	..	..	..	1 (2%)	..	..
Lower gastrointestinal haemorrhage	..	1 (2%)	..	..	..	..	..	..
Lung infection	..	1 (2%)	..	..	..	1 (2%)	..	..
Lymphocyte count decreased	..	3 (7%)	..	..	..	..	..	..
Nausea	12 (29%)	..	..	..	10 (24%)	1 (2%)	..	..
Nervous system disorders	..	..	..	..	..	1 (2%)	..	..

(Table 2 continues on next page)

	Nivolumab group (n=42)				Nivolumab plus ipilimumab group(n=42)			
	Grade 1-2	Grade 3	Grade 4	Grade 5	Grade 1-2	Grade 3	Grade 4	Grade 5
(Continued from previous page)								
Neoplasms (benign, malignant, and unspecified)	..	..	..	2 (5%)	..	..	..	2 (5%)
Neutrophil count decreased	..	..	..	..	..	..	..	..
Pain	23 (55%)	2 (5%)	..	..	22 (52%)	3 (7%)	..	..
Platelet count decreased	..	1 (2%)	..	..	..	..	..	..
Pleural effusion	..	2 (5%)	..	..	..	..	..	..
Pruritus	..	..	..	..	4 (10%)	..	..	..
Pulmonary oedema	..	..	..	..	..	..	1 (2%)	..
Rash	4 (10%)	..	..	..	8 (19%)	..	..	..
Respiratory failure	..	..	..	2 (5%)	..	..	..	2 (5%)
Sepsis	..	..	..	..	..	..	..	1 (2%)
Skin infection	..	..	..	..	..	1 (2%)	..	..
Small intestinal obstruction	..	1 (2%)	..	..	..	1 (2%)	..	..
Spinal fracture	..	1 (2%)	..	..	..	..	..	..
Thromboembolic event	..	..	..	1 (2%)	..	1 (2%)	..	..
Urinary tract infection	..	1 (2%)	..	..	..	3 (7%)	..	..
Urinary tract obstruction	..	2 (5%)	..	..	..	..	..	..
Urine output decreased	..	..	1 (2%)	..	..	..	..	..
Vaginal fistula	..	..	..	..	..	1 (2%)	..	..
Vomiting	..	1 (2%)	..	..	..	..	..	..

Adverse events are reported in all patients who initiated treatment (n=84). Adverse events are reported as per CTCAE version 4.0. Grade 1-2 adverse events are listed if reported by at least 10% of patients. All grade 3, 4 and 5 adverse events are reported. CTCAE=Common Terminology Criteria for Adverse Events.

Table 2: Adverse events by treatment group (safety population)

treatment-related adverse events reported here for nivolumab plus ipilimumab is comparable to that of chemotherapeutic agents in this setting.²⁻⁶

A limitation of this study is that it was not designed to directly compare safety and activity between the two treatment groups. Radiographical assessments did not include evaluation by immune-related response criteria or immune-related RECIST. The small sample sizes precluded stratification by relevant baseline characteristics. Research biopsies were not mandatory and therefore comprehensive analysis of biomarkers and the tumour microenvironment, and correlations of clinical activity with changes in the tumour microenvironment, were not possible. Paired tumour samples were collected from 22 patients, and correlative analysis, including PD-L1 expression and tumour-infiltrating lymphocyte characterisation by immuno-histochemistry, and mutational burden, neoantigen analysis, T-cell receptor clonality by sequencing are ongoing and will be reported in a future publication. Additionally, a trial expansion has been approved to

further characterise the activity of nivolumab and nivolumab plus ipilimumab in patients with undifferentiated pleomorphic sarcoma and liposarcoma. Enrolment had not yet begun for the expansion at the time of writing.

There is a clear need to develop predictive biomarkers to establish which patients with sarcoma are most likely to benefit from checkpoint blockade. Several studies in melanoma and lung cancers have shown that patients with a high tumour mutational load benefit from treatment with checkpoint blocking agents.^{28,29} More recently, in a study of patients with urothelial cancer treated with atezolizumab, the presence of liver metastases and low peripheral T-cell clonality seemed to inversely correlate with survival.⁸ Additionally, the presence of tumour-infiltrating lymphocytes and expression of PD-L1 in these lymphocytes have also been described as being potentially important biomarkers to predict benefit in selected malignancies.³⁰ In sarcoma, the expression of genes related to antigen presentation and T-cell infiltration is higher in undifferentiated pleomorphic sarcoma and leiomyosarcoma than in synovial sarcoma and liposarcoma.²⁰ The activity observed in this trial with nivolumab plus ipilimumab in patients with both, undifferentiated pleomorphic sarcoma and leiomyosarcoma, complement these preclinical findings. The clinical activity of the combination therapy in patients with leiomyosarcoma contrasts with what has been previously reported in this setting with nivolumab or pembrolizumab monotherapies.^{19,31} Ipilimumab might increase T-cell activation, thereby allowing nivolumab to augment T-cell responses against tumour cells. We postulate that the baseline antitumour immunity of patients with some sarcoma subtypes might be maximised with checkpoint inhibitor combinations. Unfortunately, data are scarce regarding the tumour immune microenvironment composition in most subtypes of sarcoma and this hypothesis will need to be prospectively tested in a clinical trial that encompasses robust correlative studies.

A future phase 3 clinical trial should be planned in patients with specific sarcoma subtypes that show baseline infiltration of tumour-infiltrating lymphocytes and those subtypes in which checkpoint inhibitors have clinical activity, such as undifferentiated pleomorphic sarcoma or malignant fibrous histiocytoma, leiomyosarcoma, angiosarcoma, and myxofibrosarcoma. In addition, enrolling patients earlier in their treatment course might further enhance responses. This effect has been seen in patients with Merkel cell carcinoma treated with avelumab, whereby chemotherapy-refractory patients had fewer objective responses than chemotherapy-naïve patients.¹⁷

In conclusion, in this study treatment with nivolumab plus ipilimumab in an unselected cohort of heavily treated patients with advanced sarcoma, achieved a

proportion of 16% of 38 patients with confirmed objective responses, which is similar to the results obtained with standard chemotherapy. Compared with patients who received nivolumab monotherapy, those who received nivolumab plus ipilimumab met the predefined activity endpoint, thereby warranting further study. These clinical results highlight the promise of combined checkpoint inhibition in this setting, showing a path forwards for future studies in sarcoma.

Contributors

SPD'A, MRM, GKS, WDT, and HS led the study design with support from Alliance and NCI-CTEP. Alliance and MRM did the data management. SPD'A and MRM did the data monitoring and reporting. SPD'A, MRM, JA, MMM, and WDT did the data analysis. SPD'A, MRM, JA, MMM, CRA, WDT, and GKS did the data interpretation and writing. All authors were involved in the data collection and reviewed and approved the final manuscript. Additional investigators contributed to recruitment, treatment, and follow-up of patients, as shown in the appendix (p 9).

Declaration of interests

BAVT has received grant funding from Merck and Pfizer, is a consultant to Janssen, Eli Lilly, Novartis, and Karyopharm, and Caris, and is a speaker for Eli Lilly, Janssen, and Caris. EH is an employee of Astellas (as of August, 2016). WDT has received personal fees from Eli Lilly, EMD Serono, Novartis, Eisai, Janssen, Immune Design, Adaptimmune, Ariad, Daiichi Sankyo, Plexxikon, Morphotek, Advaxis, and Traccon and has patents pending for ATRX as a companion diagnostic for CDK4 inhibitors and drug discovery. All other authors declare no competing interests.

Acknowledgments

The study was supported by the Alliance Clinical Trials in Oncology Group and the National Cancer Institute-Cancer Therapy Evaluation Program (appendix p 11), as well as Bristol-Myers Squibb. Additional funding was obtained from Cycle for Survival. We also thank the patients and their families, as well as all the investigators and study teams, for their contribution to the study. Finally, we thank Tiffany R Winfield and Vera J Suman of the Alliance Clinical Trials in Oncology Group.

References

- 1 Siegel RL, Miller KD, Jemal A. Cancer statistics, 2016. *CA Cancer J Clin* 2016; **66**: 7–30.
- 2 Seddon B, Strauss SJ, Whelan J, et al. Gemcitabine and docetaxel versus doxorubicin as first-line treatment in previously untreated advanced unresectable or metastatic soft-tissue sarcomas (GeDDiS): a randomised controlled phase 3 trial. *Lancet Oncol* 2017; **18**: 1397–410.
- 3 Demetri GD, von Mehren M, Jones RL, et al. Efficacy and safety of trabectedin or dacarbazine for metastatic liposarcoma or leiomyosarcoma after failure of conventional chemotherapy: results of a phase III randomized multicenter clinical trial. *J Clin Oncol* 2016; **34**: 786–93.
- 4 Schöffski P, Chawla S, Maki RG, et al. Eribulin versus dacarbazine in previously treated patients with advanced liposarcoma or leiomyosarcoma: a randomised, open-label, multicentre, phase 3 trial. *Lancet* 2016; **387**: 1629–37.
- 5 van der Graaf WT, Blay JY, Chawla SP, et al. Pazopanib for metastatic soft-tissue sarcoma (PALETTE): a randomised, double-blind, placebo-controlled phase 3 trial. *Lancet* 2012; **379**: 1879–86.
- 6 Tap WD, Jones RL, Van Tine BA, et al. Olaratumab and doxorubicin versus doxorubicin alone for treatment of soft-tissue sarcoma: an open-label phase 1b and randomised phase 2 trial. *Lancet* 2016; **388**: 488–97.
- 7 Hammers H, Plimack ER, Infante JR, et al. Phase I study of nivolumab in combination with ipilimumab in metastatic renal cell carcinoma. *Ann Oncol* 2014; **25** (suppl): iv361–62 (abstr).
- 8 Rosenberg JE, Hoffman-Censits J, Powles T, et al. Atezolizumab in patients with locally advanced and metastatic urothelial carcinoma who have progressed following treatment with platinum-based chemotherapy: a single-arm, multicentre, phase 2 trial. *Lancet* 2016; **387**: 1909–20.
- 9 Weber JS, D'Angelo SP, Minor D, et al. Nivolumab versus chemotherapy in patients with advanced melanoma who progressed after anti-CTLA-4 treatment (CheckMate 037): a randomised, controlled, open-label, phase 3 trial. *Lancet Oncol* 2015; **16**: 375–84.
- 10 Wolchok JD, Kluger H, Callahan MK, et al. Nivolumab plus ipilimumab in advanced melanoma. *N Engl J Med* 2013; **369**: 122–33.
- 11 Heery CR, O'Sullivan-Coyne G, Madan RA, et al. Avelumab for metastatic or locally advanced previously treated solid tumours (JAVELIN Solid Tumor): a phase 1a, multicohort, dose-escalation trial. *Lancet Oncol* 2017; **18**: 587–98.
- 12 Herbst RS, Baas P, Kim DW, et al. Pembrolizumab versus docetaxel for previously treated, PD-L1-positive, advanced non-small-cell lung cancer (KEYNOTE-010): a randomised controlled trial. *Lancet* 2016; **387**: 1540–50.
- 13 Hodi FS, O'Day SJ, McDermott DF, et al. Improved survival with ipilimumab in patients with metastatic melanoma. *N Engl J Med* 2010; **363**: 711–23.
- 14 Pardoll DM. The blockade of immune checkpoints in cancer immunotherapy. *Nat Rev Cancer* 2012; **12**: 252–64.
- 15 D'Angelo SP, Shoushtari AN, Agaram NP, et al. Prevalence of tumor-infiltrating lymphocytes and PD-L1 expression in the soft tissue sarcoma microenvironment. *Hum Pathol* 2015; **46**: 357–65.
- 16 Raj S, Bui M, Gonzales R, Letson D, Antonia SJ. Impact of PD-L1 expression on clinical outcomes in subtypes of sarcoma. *Ann Oncol* 2014; **25** (suppl 4): iv498 (abstract).
- 17 Kaufman HL, Russell J, Hamid O, et al. Avelumab in patients with chemotherapy-refractory metastatic Merkel cell carcinoma: a multicentre, single-group, open-label, phase 2 trial. *Lancet Oncol* 2016; **17**: 1374–85.
- 18 Maki RG, Jungbluth AA, Gnjatich S, et al. A pilot study of anti-CTLA4 antibody ipilimumab in patients with synovial sarcoma. *Sarcoma* 2013; **2013**: 168145.
- 19 Tawbi HA, Burgess MA, Bolejack V, et al. Pembrolizumab in advanced soft-tissue sarcoma and bone sarcoma (SARC028): a multicentre, two-cohort, single-arm, open-label, phase 2 trial. *Lancet Oncol* 2017; **18**: 1493–1501.
- 20 Pollack SM, He Q, Yearley JH, et al. T-cell infiltration and clonality correlate with programmed cell death protein 1 and programmed death-ligand 1 expression in patients with soft tissue sarcomas. *Cancer* 2017; **123**: 3291–304.
- 21 Curran MA, Montalvo W, Yagita H, Allison JP. PD-1 and CTLA-4 combination blockade expands infiltrating T cells and reduces regulatory T and myeloid cells within B16 melanoma tumors. *Proc Natl Acad Sci U S A* 2010; **107**: 4275–80.
- 22 Hellmann MD, Rizvi NA, Goldman JW, et al. Nivolumab plus ipilimumab as first-line treatment for advanced non-small-cell lung cancer (CheckMate 012): results of an open-label, phase 1, multicohort study. *Lancet Oncol* 2017; **18**: 31–41.
- 23 Pocock SJ, Simon R. Sequential treatment assignment with balancing for prognostic factors in the controlled clinical trial. *Biometrics* 1975; **31**: 103–15.
- 24 Simon R. Optimal two-stage designs for phase II clinical trials. *Control Clin Trials* 1989; **10**: 1–10.
- 25 Kaplan EL, Meier P. Nonparametric-estimation from incomplete observations. *J Am Stat Assoc* 1958; **53**: 457–81.
- 26 Wolchok JD, Neyns B, Linette G, et al. Ipilimumab monotherapy in patients with pretreated advanced melanoma: a randomised, double-blind, multicentre, phase 2, dose-ranging study. *Lancet Oncol* 2010; **11**: 155–64.
- 27 Postow MA, Chesney J, Pavlick AC, et al. Nivolumab and ipilimumab versus ipilimumab in untreated melanoma. *N Engl J Med* 2015; **372**: 2006–17.
- 28 Rizvi NA, Hellmann MD, Snyder A, et al. Cancer immunology. Mutational landscape determines sensitivity to PD-1 blockade in non-small cell lung cancer. *Science* 2015; **348**: 124–28.
- 29 Snyder A, Makarov V, Merghoub T, et al. Genetic basis for clinical response to CTLA-4 blockade in melanoma. *N Engl J Med* 2014; **371**: 2189–99.
- 30 Tumeh PC, Harview CL, Yearley JH, et al. PD-1 blockade induces responses by inhibiting adaptive immune resistance. *Nature* 2014; **515**: 568–71.
- 31 Ben-Ami E, Barysaukas CM, Solomon S, et al. Immunotherapy with single agent nivolumab for advanced leiomyosarcoma of the uterus: results of a phase 2 study. *Cancer* 2017; **123**: 3285–90.